

Exercise JA

Q1. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation

$$(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$$

Where $a > -1$, then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are :

(A) $-\frac{5}{2}$ & 1 (B) $-\frac{5}{2}$ & -1 (C) $-\frac{7}{2}$ & 2 (D) $-\frac{9}{2}$ & 3

Ans. B

Sol. -

Q2. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$ then

(A) $a = 1, b = 4$ (B) $a = 1, b = -4$ (C) $a = 2, b = -3$ (D) $a = 2, b = 3$

Ans. B

Sol. -

Q3. The largest value of the nonnegative integer a for which

$$\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4} \text{ is :}$$

Ans. 0

Sol. -

Q4. Let $f(x) = \sin \left(\frac{\pi}{6} \sin \left(\frac{\pi}{2} \sin x \right) \right)$ for all $x \in \mathbb{R}$ and $g(x) = \frac{\pi}{2} \sin x$ for all $x \in \mathbb{R}$. Let $(f \circ g)(x)$ denote

$f(g(x))$ and $(g \circ f)(x)$ denote $g(f(x))$. Then which of the following is (are) true ?

(A) Range of f is $\left[-\frac{1}{2}, \frac{1}{2} \right]$ (B) Range of $f \circ g$ is $\left[-\frac{1}{2}, \frac{1}{2} \right]$ (C) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{\pi}{6}$

(D) There is an $x \in \mathbb{R}$ such that $(g \circ f)(x) = 1$

Ans. A, B, C

Sol. -

Q5. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n) - e}}{\alpha^m} \right) = -\left(\frac{e}{2} \right)$. Then the value of $\frac{m}{n}$ is :

Ans. 2

Sol. -

Q6. Let $a, b \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals

Ans. 7

Sol. -

Q7. Let $f(x) = \frac{1 - x(1 + |1 - x|)}{|1 - x|} \cos\left(\frac{1}{1 - x}\right)$ for $x \neq 1$. Then :

(A) $\lim_{x \rightarrow 1^+} f(x) = 0$ (B) $\lim_{x \rightarrow 1^-} f(x) = 0$ (C) $\lim_{x \rightarrow 1^+} f(x)$ does not exist

(D) $\lim_{x \rightarrow 1^-} f(x)$ does not exist

Ans. C, B

Sol. -

Q8. Let e denote the base of the natural logarithm. The value of the real number a for which the right hand limit $\lim_{x \rightarrow 0^+} \frac{(1 - x)^{\frac{1}{x}} - e^{-1}}{x^a}$ is equal to a nonzero real number, is

Ans. 1

Sol.

$$\lim_{x \rightarrow 0^+} \frac{(1 - x)^{\frac{1}{x}} - e^{-1}}{x^a}$$

$$\lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x} \ln(1-x)} - e^{-1}}{x^a} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{e^{-1}(e^{1 + \frac{1}{x} \ln(1-x)} - 1)}{x^a} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{1 + \frac{1}{x} \ln(1-x)}{ex^a} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{x + \ln(1-x)}{ex^{a+1}} \neq 0$$

$$\lim_{x \rightarrow 0^+} \frac{x + \left(-x - \frac{x^2}{2} \dots\right)}{ex^{a+1}} \neq 0$$

$\Rightarrow a + 1 = 2$

$\Rightarrow a = 1$

Q9. The value of the limit $\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2\sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2}\cos 2x + \cos \frac{3x}{2}\right)}$

Ans. 8

Sol.

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin 3x + \sin x)}{\left(2\sin 2x \sin \frac{3x}{2} + \cos \frac{5x}{2}\right) - \left(\sqrt{2} + \sqrt{2}\cos 2x + \cos \frac{3x}{2}\right)}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2(2 \sin 2x \cos x)}}{\left[\cos \frac{x}{2} - \cos \frac{7x}{2} + \cos \frac{5x}{2} \right] - \left[\sqrt{2}(2 \cos^2 x) + \cos \frac{3x}{2} \right]}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(2 \sin 2x \cos x)}{\cos \frac{x}{2} - \cos \frac{3x}{2} + \cos \frac{5x}{2} - \cos \frac{7x}{2} - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin \cos^2 x}{2 \sin x \sin \frac{x}{2} + 2 \sin 3x \sin \frac{x}{2} - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin \cos^2 x}{2 \sin x \frac{x}{2} (\sin 3x + \sin x) - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin \cos^2 x}{2 \sin x \frac{x}{2} (2 \sin 2x \cos x) - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2} \sin \cos^2 x}{8 \sin \frac{x}{2} \sin x \cos^2 x - 2\sqrt{2} \cos^2 x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{16\sqrt{2}}{8 \times \frac{1}{\sqrt{2}} - 2\sqrt{2}} = 8$$

Q10. For any positive integer n, let $S_n : (0, \infty) \rightarrow R$ be defined by

$S_n(x) = \sum_{k=1}^n \cot^{-1} \left(\frac{1 + k(k+1)x^2}{x} \right)$, where for any $x \in R$, $\cot^{-1}(x) \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ Then which of the following statements is (are) TRUE ?

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1} \left(\frac{1 + 11x^2}{10x} \right)$ for all $x > 0$ (B) $\lim_{x \rightarrow \infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

Ans. A, B

Sol. $S_n(x) = \sum_{k=1}^n \cot^{-1} \left(\frac{1 + k(k+1)x^2}{x} \right)$
 $= \sum_{k=1}^n \tan^{-1} \left(\frac{(k+1)x - kx}{1 + (k+1)xkx} \right)$
 $= \sum_{k=1}^n \tan^{-1}((k+1)x) - \tan^{-1}kx$
 $= (\tan^{-1}2x - \tan^{-1}x) + (\tan^{-1}3x - \tan^{-1}2x) + \dots + (\tan^{-1}(n+1)x - \tan^{-1}nx)$
 $S_n(x) = \tan^{-1}(n+1)x - \tan^{-1}x$

(A) $S_{10}(x) = \tan^{-1} 11x - \tan^{-1} x$, $x > 0$
 $= \tan^{-1} \left(\frac{11x - x}{1 + 11x \cdot x} \right)$
 $= \tan^{-1} \left(\frac{10x}{1 + 11x^2} \right)$
 $= \cot^{-1} \left(\frac{1 + 11x^2}{10x} \right)$
 $= \frac{\pi}{2} - \tan^{-1} \left(\frac{1 + 11x^2}{10x} \right) \Rightarrow \text{(A) is correct}$

$$\begin{aligned}
\text{(B)} \quad \lim_{n \rightarrow \infty} \cot(S_n(x)) &= \lim_{n \rightarrow \infty} \cot(\tan^{-1}(n+1)x - \tan^{-1}x), x > 0 \\
&= \lim_{n \rightarrow \infty} \cot\left(\tan^{-1} \frac{(n+1)x - x}{1 + (n+1)x \cdot x}\right) \\
&= \lim_{n \rightarrow \infty} \cot\left(\tan^{-1} \frac{nx}{1 + (n+1)x^2}\right) \\
&= \lim_{n \rightarrow \infty} \cot\left[\tan^{-1} \frac{x}{\frac{1}{n} + \left(1 + \frac{1}{n}\right)x^2}\right] = \cot \tan^{-1}\left(\frac{x}{x^2}\right) \\
&= \cot(\cot^{-1}x) = x
\end{aligned}$$

$\Rightarrow$ (B) is correct

(C) $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

$$\tan^{-1} 4x - \tan^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow \frac{3x}{1+4x^2} = 1 \Rightarrow 4x^2 + 1 = 3x \Rightarrow 4x^2 - 3x + 1 = 0$$

$D < 0$

$\Rightarrow$ no real

$\Rightarrow$ (C) option is incorrect

(D) $S_n(x) = \tan^{-1}(n+1)x - \tan^{-1}x$

$$= \tan^{-1}\left(\frac{nx + x - x}{1 + (n+1)x^2}\right)$$

$$= \tan^{-1}\left(\frac{nx}{1 + (n+1)x^2}\right)$$

$$\tan(S_n(x)) = \frac{nx}{1 + (n+1)x^2} = \frac{n}{\frac{1}{x} + (n+1)x}$$

$\Rightarrow A.M. \geq G.M.$

$$\frac{\frac{1}{x} + (n+1)x}{2} \geq \sqrt{\frac{1}{x} \cdot (n+1)x}$$

$$\frac{1}{x} + (n+1)x \geq 2\sqrt{n+1}$$

$$\Rightarrow \tan(S_n(x)) = \frac{nx}{1 + (n+1)x^2} \leq \frac{n}{2\sqrt{n+1}}$$

for $n = 2$

$$\frac{n}{2\sqrt{n+1}} = \frac{1}{\sqrt{3}} > \frac{1}{2}$$

D is incorrect

Q11. If $\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1 - x^3)^{\frac{1}{3}} + \left((1 - x^2)^{\frac{1}{2}} - 1\right) \sin x}{x \sin^2 x}$ then the value of 6β is ____.

Ans. 5

Sol.

$$\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1 - x^3)^{1/3}}{\frac{x \sin^2 x}{x^2} x^2} + \frac{\left((1 - x^2)^{1/2} - 1\right) \sin x}{x \frac{\sin^2 x}{x^2} x^2}$$

use expansion

Q12. Let $k \in \mathbb{R}$. If $\lim_{x \rightarrow 0+} (\sin(\sin kx) + \cos x + x)^{\frac{2}{x}} = e^6$, then the value of k is :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $\lim_{x \rightarrow 0^+} (\sin(\sin kx) + \cos x + x)^{\frac{2}{x}} = e^6$

$$\lim_{x \rightarrow 0^+} 2 \left\{ \frac{\sin(\sin kx) + \cos x + x - 1}{x} \right\} = e^6$$

$$\lim_{x \rightarrow 0^+} \frac{\sin(\sin kx) + \cos x + x - 1}{x} = 3$$

$$\lim_{x \rightarrow 0^+} \frac{\cos(\sin kx) \cdot \cos(kx) \cdot k - \sin x + 1}{1} = 3$$

$$k + 1 = 3 \Rightarrow k = 2$$

Q13. Let S be the set of all $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ such that $\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\log_e x)^\alpha \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\log_e(1+x))^\beta} = 0$. Then which of the following is (are) correct?

(A) $(-1, 3) \in S$ (B) $(-1, 1) \in S$ (C) $(1, -1) \in S$ (D) $(1, -2) \in S$

Ans. B, C

Sol. $\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\ln(x))^\alpha \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\ln(1+x))^\beta} = 0$

$$\lim_{x \rightarrow \infty} \frac{\sin(x^2)[\ln(x)]^\alpha \left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\ln(1+x))^\beta} = 0$$

$$\lim_{x \rightarrow \infty} \frac{\sin(x^2)(\ln x)^\alpha}{x^{\alpha\beta+2}(\ln(1+x))^\beta} = 0$$

$$\Rightarrow \alpha\beta + 2 > 0$$

or

$$\alpha\beta + 2 = 0 \text{ \& } \beta > \alpha$$

$$\alpha\beta > -2$$

$$\alpha\beta = -2 \text{ \& } \beta > \alpha$$

(A) $\alpha\beta = -3$

Incorrect

(B) $\alpha\beta = -1$

Correct

(C) $\alpha\beta = -1$

Correct

(D) $\alpha\beta = -2$ but $\alpha > \beta$

Incorrect

Exercise C1

Q1. Let $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, then $\lim_{x \rightarrow \infty} f(x)$ equals

- (A) 0 (B) - 1/2 (C) 1 (D) none of these

Ans. C

Sol. -

Q2. Let α, β be the roots of $ax^2 + bx + c = 0$, where $1 < \alpha < \beta$. Then $\frac{|ax^2 + bx + c|}{ax^2 + bx + c} = 1$ then which of the following statements is incorrect

- (A) $a > 0$ and $x_0 < 1$ (B) $a > 0$ and $x_0 > \beta$ (C) $a < 0$ and $\alpha < x_0 < \beta$
 (D) $a < 0$ and $x_0 < 1$

Ans. D

Sol. -

Q3. If $\lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a+x}(bx - \sin x)} = 1$ then the constants 'a' and 'b' are (where $a > 0$)

- (A) $b = 1, a = 36$ (B) $a = 1, b = 6$ (C) $a = 1, b = 36$ (D) $b = 1, a = 6$

Ans. A

Sol. -

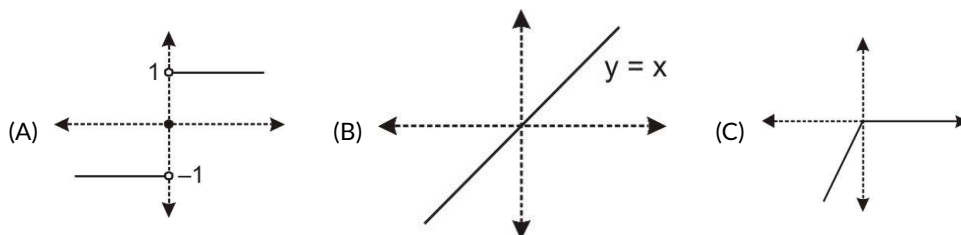
Q4. If $\ell = \lim_{x \rightarrow \infty} (\sin\sqrt{x+1} - \sin\sqrt{x})$ and $m = \lim_{x \rightarrow \infty} [\sin\sqrt{x+1} - \sin\sqrt{x}]$ where $[.]$ denotes the greatest integer function then :

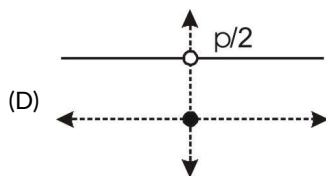
- (A) $\ell = m = 0$ (B) $\ell = 0$; m is undefined (C) ℓ, m both do not exist
 (D) $\ell = 0, m \neq 0$ (although m exist)

Ans. B

Sol. -

Q5. The graph of the function $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \cot^{-1} \frac{x}{t^2} \right)$, is





Ans. C

Sol. -

Q6. If $\lim_{x \rightarrow \infty} f(x)$ exist and is finite & nonzero and if $\lim_{x \rightarrow \infty} \left(f(x) + \frac{3f(x) - 1}{f^2(x)} \right) = 3$ then the value of $\lim_{x \rightarrow \infty} f(x)$ is

- (A) 1 (B) -1 (C) 2 (D) none of these

Ans. A

Sol. -

Q7. $\lim_{x \rightarrow \infty} \frac{e^x \left((2^{x^n})^{\frac{1}{e^x}} - (3^{x^n})^{\frac{1}{e^x}} \right)}{x^n}$, $n \in \mathbb{N}$ is equal to :

- (A) 0 (B) $\ln(2/3)$ (C) $\ln(3/2)$ (D) none

Ans. B

Sol. -

Q8. If $\lim_{x \rightarrow 0} \frac{((a - n)nx - \tan x) \sin nx}{x^2} = 0$, where n is non-zero real number, then a is equal to:

- (A) 0 (B) $\frac{n+1}{n}$ (C) n (D) $n + \frac{1}{n}$

Ans. D

Sol. -

Q9. $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is :

- (A) 2 (B) -2 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Ans. C

Sol. -

Q10. The value of $\lim_{x \rightarrow \infty} \frac{\cot^{-1}(x^{-a} \log_a x)}{\sec^{-1}(a^x \log_x a)}$ ($a > 1$) is equal to

- (A) 1 (B) 0 (C) $\pi/2$ (D) does not exist

Ans. A

Sol. -

Q11.
$$\lim_{x \rightarrow 0} \frac{\left(\sqrt{(1 - \cos x) + \sqrt{(1 - \cos x) + \sqrt{(1 - \cos x) + \dots \infty}}} \right) - 1}{x^2} \text{ equals}$$

- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

Ans. B

Sol. -

Q12.
$$\lim_{x \rightarrow \infty} x \left(\tan^{-1} \frac{x+1}{x+2} - \cot^{-1} \frac{x+2}{x} \right) \text{ is}$$

- (A) -1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) non existent

Ans. B

Sol. -

Q13. Which one of the following statement is true?

- (A) If $\lim_{x \rightarrow c} f(x) \cdot g(x)$ and $\lim_{x \rightarrow c} f(x)$ exist, then $\lim_{x \rightarrow c} g(x)$ exists.
(B) If $\lim_{x \rightarrow c} f(x) \cdot g(x)$ exists, then $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exist.
(C) If $\lim_{x \rightarrow c} (f(x) + g(x))$ and $\lim_{x \rightarrow c} f(x)$ exist, then $\lim_{x \rightarrow c} g(x)$ exist.
(D) If $\lim_{x \rightarrow c} (f(x) + g(x))$ exists, then $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exist.

Ans. C

Sol. -

Q14. If $f(x) = \cos x$, $x = n\pi$, $n = 0, 1, 2, 3, \dots$

$= 3$, otherwise and

$$\phi(x) = \begin{cases} x^2 + 1 & \text{when } x \neq 3, x \neq 0 \\ 3 & \text{when } x = 0 \\ 5 & \text{when } x = 3 \end{cases} \quad \text{then } \lim_{x \rightarrow 0} f(\phi(x)) =$$

- (A) 1 (B) 3 (C) 5 (D) None

Ans. B

Sol. -

Q15.
$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt[n]{p} + \sqrt[n]{q}}{2} \right), p, q > 0 \text{ equals}$$

- (A) 1 (B) $\sqrt{pq}$ (C) pq (D) $\frac{pq}{2}$

Ans. B

Sol. -

Q16. If $f(x) = \cot^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right)$ and $g(x) = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right)$, then $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)}$, $0 < a < \frac{1}{2}$ is :

- (A) $\frac{3}{2(1 + a^2)}$ (B) $\frac{3}{2}$ (C) $\frac{-3}{2(1 + a^2)}$ (D) $-\frac{3}{2}$

Ans. D

Sol. -

Q17. If $\lim_{x \rightarrow c^-} \{\ln x\}$ and $\lim_{x \rightarrow c^+} \{\ln x\}$ exists finitely but they are not equal (where $\{.\}$ denotes fractional part function), then :

- (A) 'c' can take only rational values (B) 'c' can take only irrational values
(C) 'c' can take infinite values in which only one is irrational
(D) 'c' can take infinite values in which only one is rational

Ans. D

Sol. -

Q18. Let $a = \lim_{x \rightarrow 1} \left(\frac{x}{\ln x} - \frac{1}{x \ln x} \right)$; $b = \lim_{x \rightarrow 0} \frac{x^3 - 16x}{4x + x^2}$; $c = \lim_{x \rightarrow 0} \frac{\ln(+\sin x)}{x}$ and $d = \lim_{x \rightarrow -1} \frac{(x+1)^3}{3[\sin(x+1) - (x+1)]}$, then the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is :

- (A) Idempotent (B) Involutary (C) Non-singular (D) Nilpotent

Ans. D

Sol. -

Q19.

$$f(x) = \begin{cases} \frac{\max\left(x, \frac{1}{x}\right)}{\min\left(x, \frac{1}{x}\right)}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}, \text{ then}$$

Consider the function $f(x)$

$$\lim_{x \rightarrow 0^-} \{f(x)\} + \lim_{x \rightarrow 1^-} \{f(x)\} + \lim_{x \rightarrow -1^-} [f(x)] =$$

(where $\{.\}$ denotes fraction part function $[.]$ denotes greatest integer function)

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. -

Q20. Let $\alpha = \lim_{n \rightarrow \infty} \frac{(1^3 - 1^2) + (2^3 - 2^2) + \dots + (n^3 - n^2)}{n^4}$, then α is equal to

- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) non existent

Ans. B

Sol. -

Q21. If $\lim_{x \rightarrow 0} [1 + x \ln(1 + b^2)]^{\frac{1}{x}} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is :

- (A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$

Ans. D

Sol. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{(1-a)x^2 + x(1-a-b) + 1-b}{x+1} \right) = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{((1-a) \cdot x + 1 - a - b + (\frac{1-b}{x}))}{1 + \frac{1}{x}} = 4$$

for limit to exist finitely

$$1 - a = 0 \text{ and } 1 - a - b = 4$$

$$\Rightarrow a = 1 \text{ and } b = -4.$$

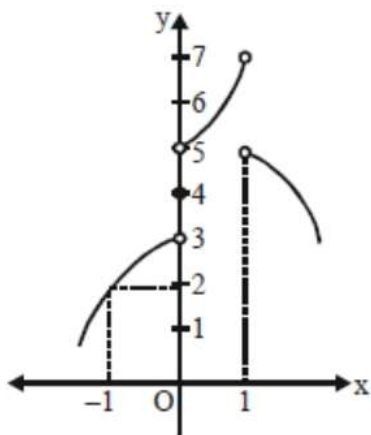
Q22. Let $f(x) = \begin{cases} |x|, & -2 \leq x \leq 0 \\ e^x - 1, & 0 < x \leq 5 \end{cases}$, $g(x) = \begin{cases} -x - 1, & -2 \leq x \leq 0 \\ \frac{4x-3}{3}, & 0 < x \leq 3 \end{cases}$ then –

- (A) $\lim_{x \rightarrow 0} f(g(x)) = 1$ (B) $\lim_{x \rightarrow 0^-} f(g(2^x)) = 1$ (C) $\lim_{x \rightarrow 0} f(g(2^x)) = e^{1/2} - 1$
 (D) $\lim_{x \rightarrow 0^+} f(g(2^x - 1)) = \lim_{x \rightarrow 0^-} f(g(2^x - 1))$

Ans. D, C, A

Sol. -

Q23. The graph of $y = f(x)$ is as shown. The which of the following hold(s) good ?
 [Note : $[k]$ and $\{k\}$ denote largest integer less than or equal k and fractional part of k respectively] –



- (A) $\lim_{x \rightarrow 0^+} f([x - \sin x]) = 4$ (B) $\lim_{x \rightarrow 0^-} f([x - \sin x]) = 2$ (C) $\lim_{x \rightarrow 0^+} \left(\left\{ \frac{x}{\tan x} \right\} \right) = 7$
 (D) $\lim_{x \rightarrow \frac{\pi}{2}^+} f(\{\cos x\}) = 5$

Ans. A, C

Sol. -

Q24. Let $f(x) = \frac{\cos^{-1}(\cos x)}{x}$, then which of the following is (are) correct (where $[.]$ denotes greatest integer function) ?

- (A) $\lim_{x \rightarrow 0^+} [f(x)] = 1$ (B) $\lim_{x \rightarrow 0^-} [f(x)] = 1$ (C) $\lim_{x \rightarrow 0^+} [f(x)] = 0$ (D) $\lim_{x \rightarrow 0^-} [f(x)] = -1$

Ans. D, A

Sol. -

Q25. Which of the following limits does not exist ?

- (A) $\lim_{x \rightarrow 1^+} ([x])^{\frac{1}{x-1}}$ (B) $\lim_{x \rightarrow 3} \frac{(x^2 - 9 - \sqrt{x^2 - 6x + 9})}{|x - 1| - 2}$ (C) $\lim_{x \rightarrow 1^+} (x)^{\ln x}$
 (D) $\lim_{x \rightarrow 0^+} \left(\frac{1 - \cot(\sin^2 x)}{x^2} \right)^{\frac{\ln(1-2x^2)}{\sin^2 x}}$

Ans. B, C, D

Sol. -

Q26. Which of the following limits are non existent ?

- (A) $\lim_{x \rightarrow 0} \frac{1}{x} \sqrt{\frac{1 - \cos x}{1 + \cos x}}$ (B) $\lim_{x \rightarrow 0} \frac{|x|}{x}$ (C) $\lim_{x \rightarrow 0} \frac{\cot^{-1} \frac{1}{x}}{x}$
 (D) $\lim_{x \rightarrow \pi/2} \cos^{-1} [\cot x]$ where $[]$ is greatest integer function

Ans. A, B, C, D

Sol. -

Q27.

Let $f(x) = \begin{cases} \frac{x}{\tan x} & , -\infty < x < 0 \\ 2 & , x = 0 \\ \frac{\ln(1+2x)}{x} & , 0 < x < \infty \end{cases}$ and $g(x) = \begin{cases} x+4 & , -\infty < x < 1 \\ x^2 - 5x + 11, & 1 \leq x < 2 \\ x-3 & , 2 \leq x < \infty \end{cases}$, then which of the following is(are) correct ?

- (A) $\lim_{x \rightarrow 0^-} g(f(x)) = 5$ (B) $\lim_{x \rightarrow 0^-} g(f(x)) = 7$ (C) $\lim_{x \rightarrow 0^+} g(f(x)) = 5$ (D) $\lim_{x \rightarrow 0^+} g(f(x)) = -1$

Ans. C, A

Sol. -

Q28. Let $f(x) = \begin{cases} |x-2| + a^2 - 6a + 9, & x < 2 \\ 5 - 2x & x \geq 2 \end{cases}$

If $\lim_{x \rightarrow 2} [f(x)]$ exists, the possible values a can take is/are (where $[.]$ represents the greatest integer function)

- (A) 2 (B) $\frac{5}{2}$ (C) 3 (D) $\frac{7}{2}$

Ans. D, C, B

Sol. -

Q29. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then

- (A) $a = 2$ (B) $a = 1$ (C) $L = \frac{1}{64}$ (D) $L = \frac{1}{32}$

Ans. A, C

Sol. -